

REAL ESTATE

Tuck Business Bridge Elective

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About me

- Education

- Bachelor's and master's degrees in philosophy
- PhD in economics

- Work (pre-PhD)

- Investment research analyst at hedge fund

- Teaching (Tuck MBAs)

- Real Estate
- Housing

- Research

- Household finance

Class plan

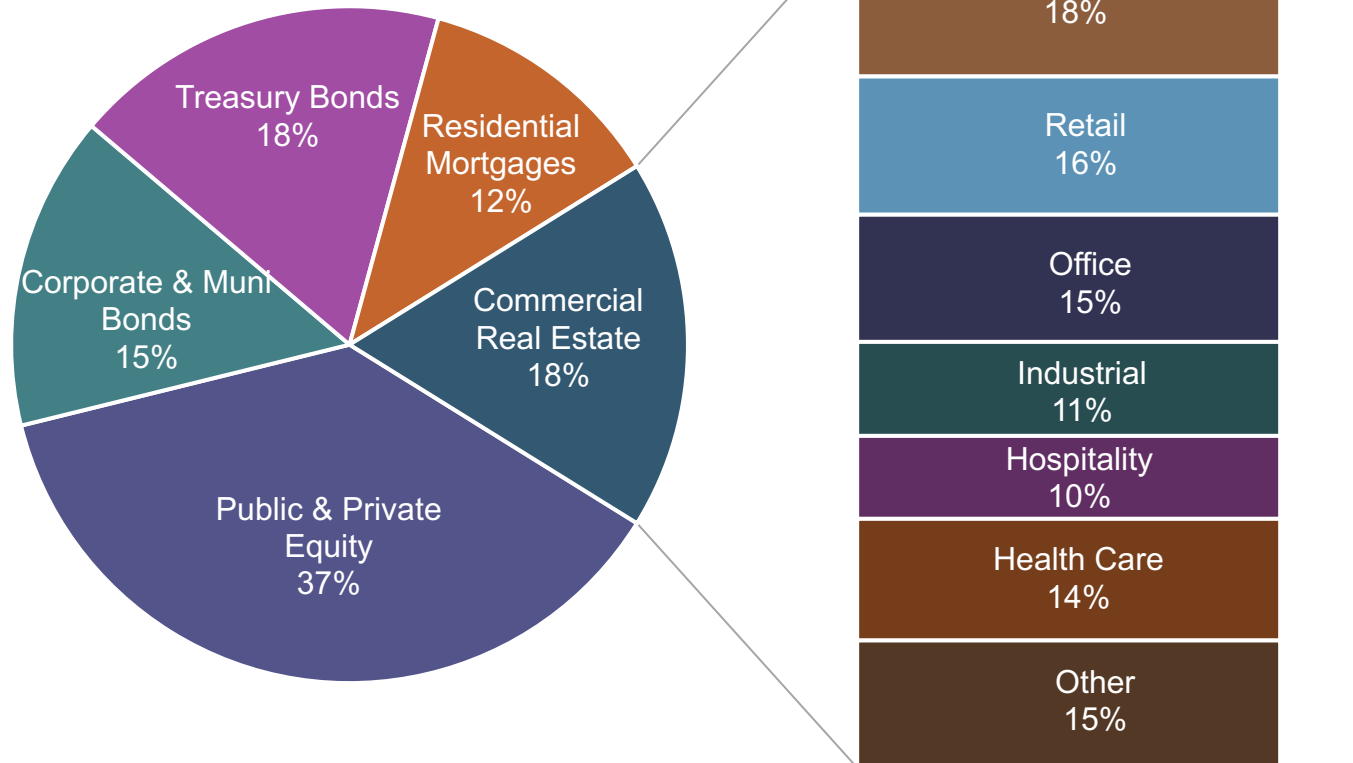
- Provide overview of real estate sector
 - Basic economics of real estate
 - Investing in real estate
- Case study on an investment decision

Why Real Estate?

- Real estate is fundamental – people need places to live, work, shop, recreate, etc.
- Real estate is a large asset class for **investors**
- Real estate lending is important to **banks**

Real estate as an asset class

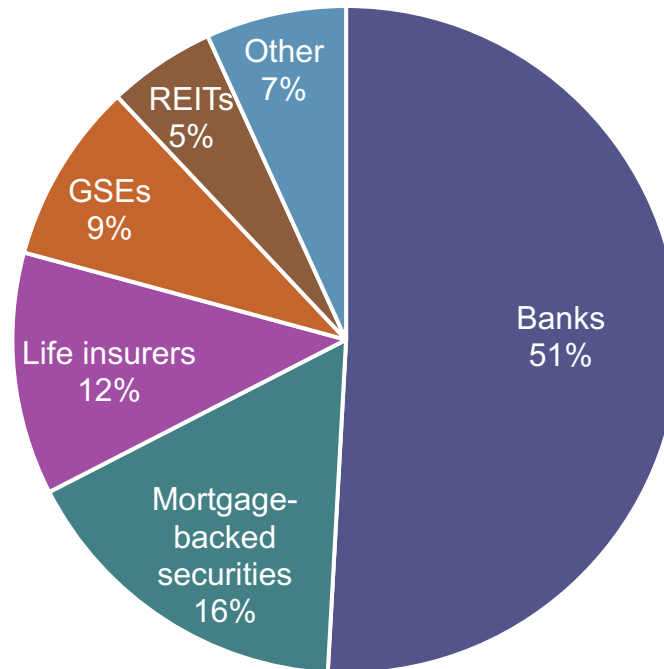
- Commercial real estate comprises large share of asset market



Sources: Nareit, SIFMA, Wilshire, Federal Reserve Board

Commercial real estate (CRE) debt

- Banks hold the majority of commercial real estate debt (June 2023)



- CRE debt accounts for 25% of bank lending (46% for small banks)

Sources: Federal Reserve Board Z.1, FRB St. Louis

AVB multifamily apartments

OPERATING A HIGH-QUALITY, DIVERSE PORTFOLIO

GARDEN



AVALON HUNTINGTON STATION
HUNTINGTON STATION, NY

MID-RISE



AVA WHEATON
SILVER SPRING, MD

HIGH-RISE



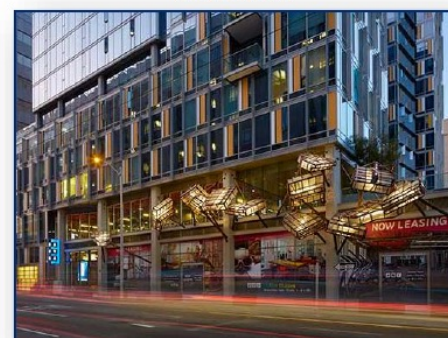
AVALON NORTH STATION
BOSTON, MA



EAVES RANCHO PENASQUITOS
SAN DIEGO, CA



AVALON WEST HOLLYWOOD
HOLLYWOOD, CA



AVA 55 NINTH
SAN FRANCISCO, CA

The Company defines urban submarkets as those with more than 3,500 households per square mile; all other submarkets are considered suburban.

STAG Industrial warehouse



© VHT STUDIOS

DEMAND-SUPPLY DYNAMICS IN REAL ESTATE

Identifying good real estate investments

- Would you invest in data centers?
- How about a condominium in Vail, CO?

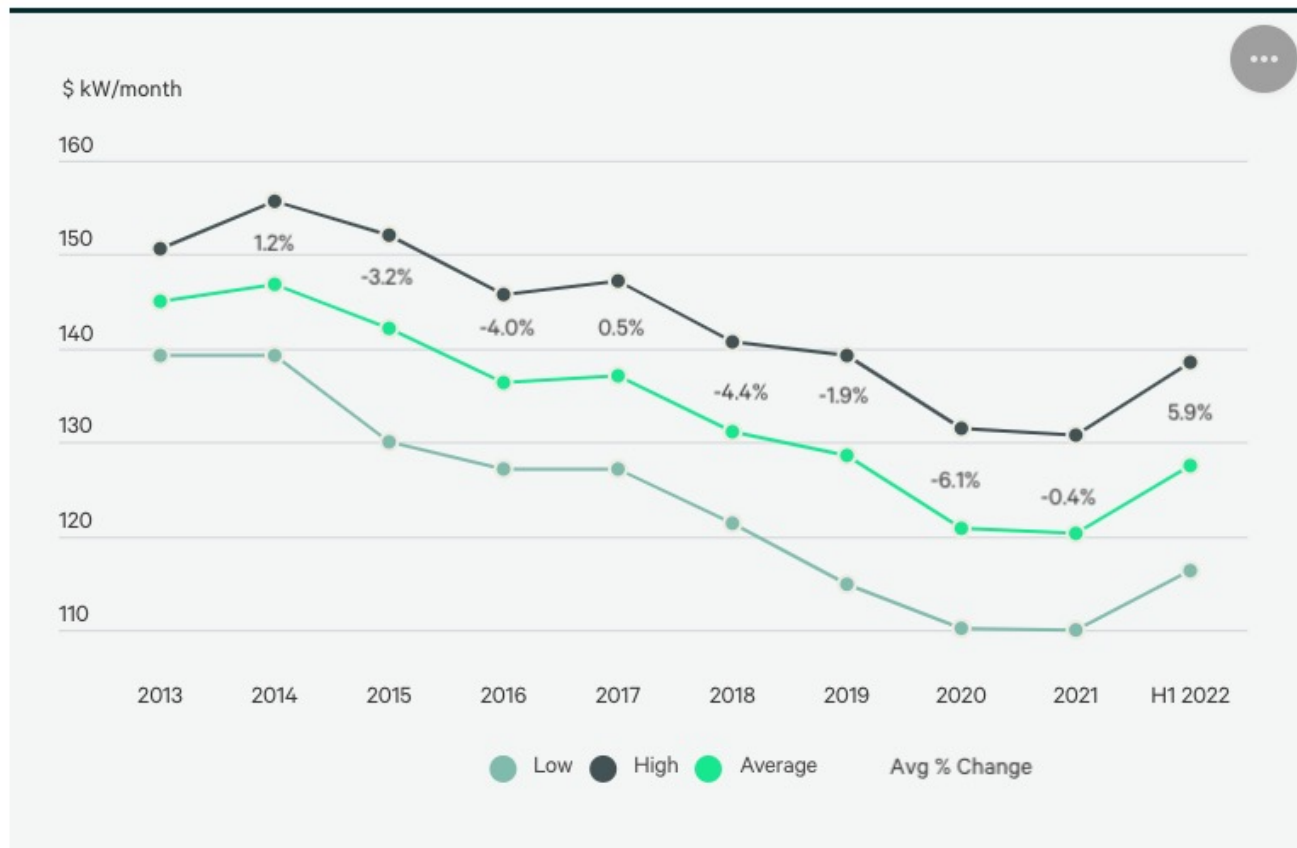
Short-run equilibrium

Long-run equilibrium

Demand-supply dynamics

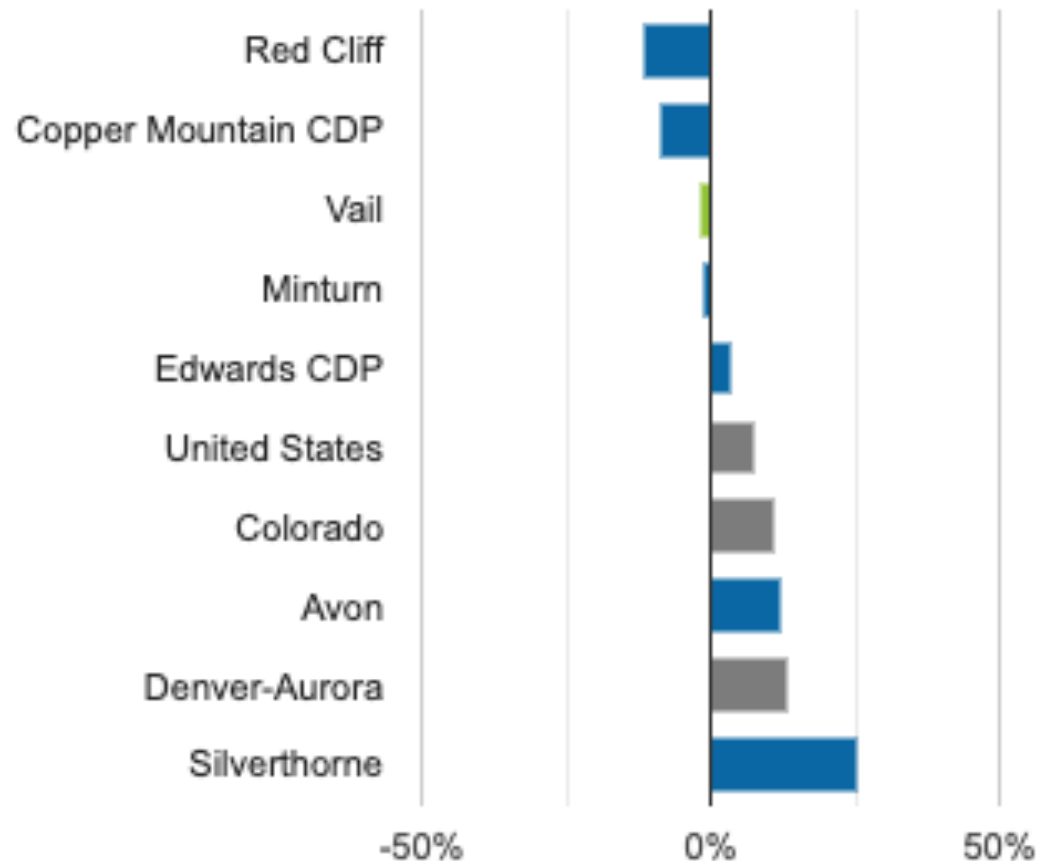
Data center rents (CBRE)

Figure 7: Average Asking Rental Rate with Y-o-Y % Change for Primary Markets

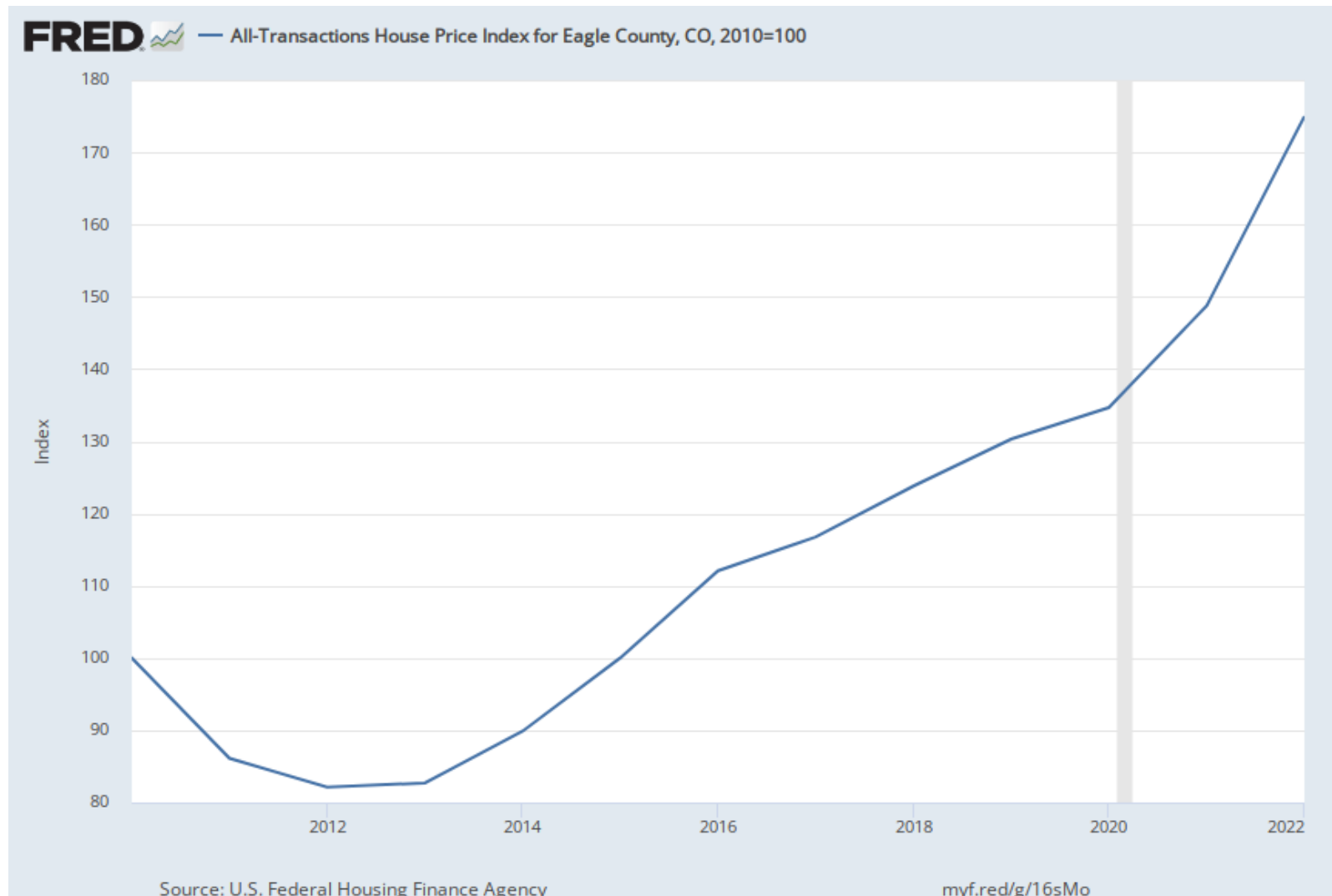


Vail – housing supply growth

Figure 4: Change in Housing Units from 2010 to 2021



Vail (Eagle Cnty) home price index



Recent demand drivers

- Secular shift toward online shopping
 - ↓ demand for retail space
 - ↑ demand for industrial space
- Post-COVID remote work
 - ↓ demand for office space
 - ↑ demand for housing
 - ↑ demand for suburbs and “Zoom towns” vs. cities

Strategy – VRIN framework

- **Value** accrues to those who own **rare** and **inimitable** resource that is in demand
- In real estate that resource is typically land (location)
- Supply constraints are critical
 - Physical constraints (limits on available land)
 - Regulatory constraints (limits on development)

Why Real Estate?

- Real estate careers draw on various skills
 - Financial and economic analysis
 - Architecture and design
 - Project management in development
 - Sales and marketing

Testimony

“Looking back, I can only wish I had become a real estate investment manager...

Real estate is fascinating. To a far greater degree than stocks and bonds, real estate combines sociology, geography, demography, architecture, and political forces, with the dynamics of fundamental economic trends, complex financing problems, the perils of illiquidity, and subtle valuation considerations.”

- Peter Bernstein, “Real Estate Matters,” *Journal of Portfolio Management*, September 2003

FINANCIAL ANALYSIS OF REAL ESTATE INVESTMENTS

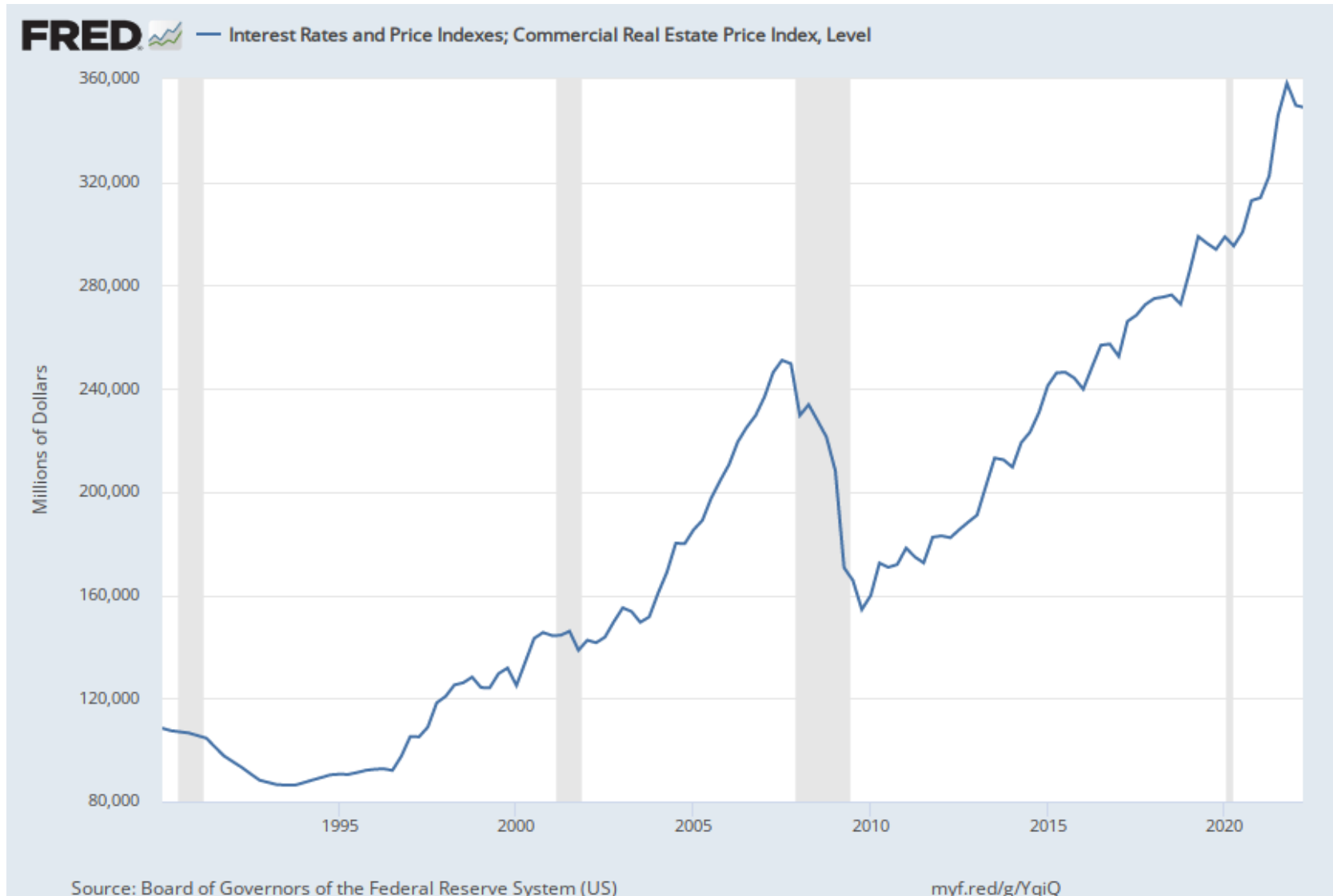
Risk and return in real estate

- Are real estate cash flows...
 - Volatile?
 - Cyclical?
 - Sensitive to inflation?
 - Correlated with financial market returns?

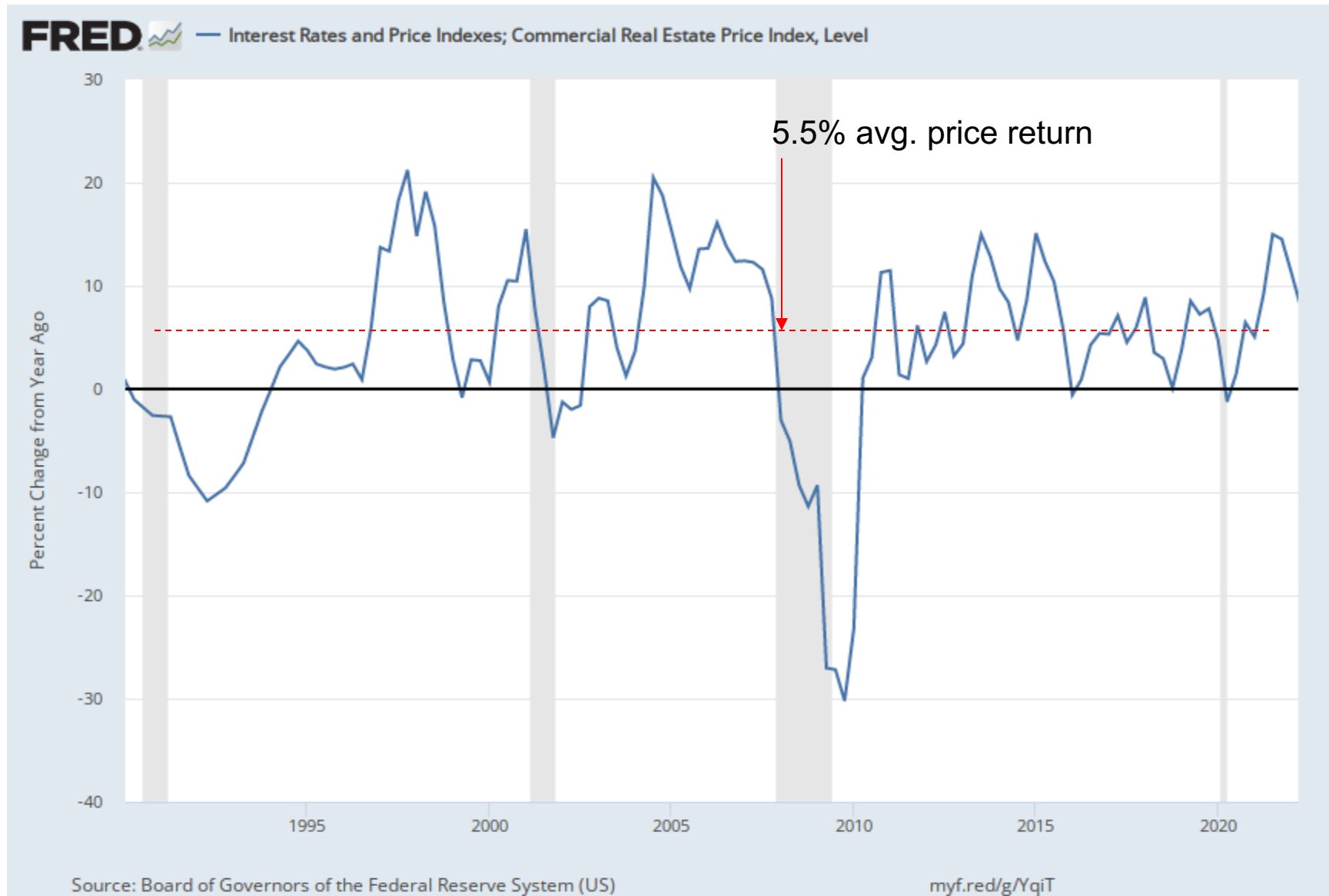
Characterizing real estate investments

- Long-term lease contracts provide **cash flow predictability**
 - Contractual price escalation provides **inflation hedge**
- Real estate sectors vary in cash flow **cyclicality**
 - Hotel cash flows are quite cyclical whereas apartment cash flows are quite stable.
- Real estate properties are **illiquid** and **costly to trade**
 - Brokerage commissions
 - Lengthy timeline to sale

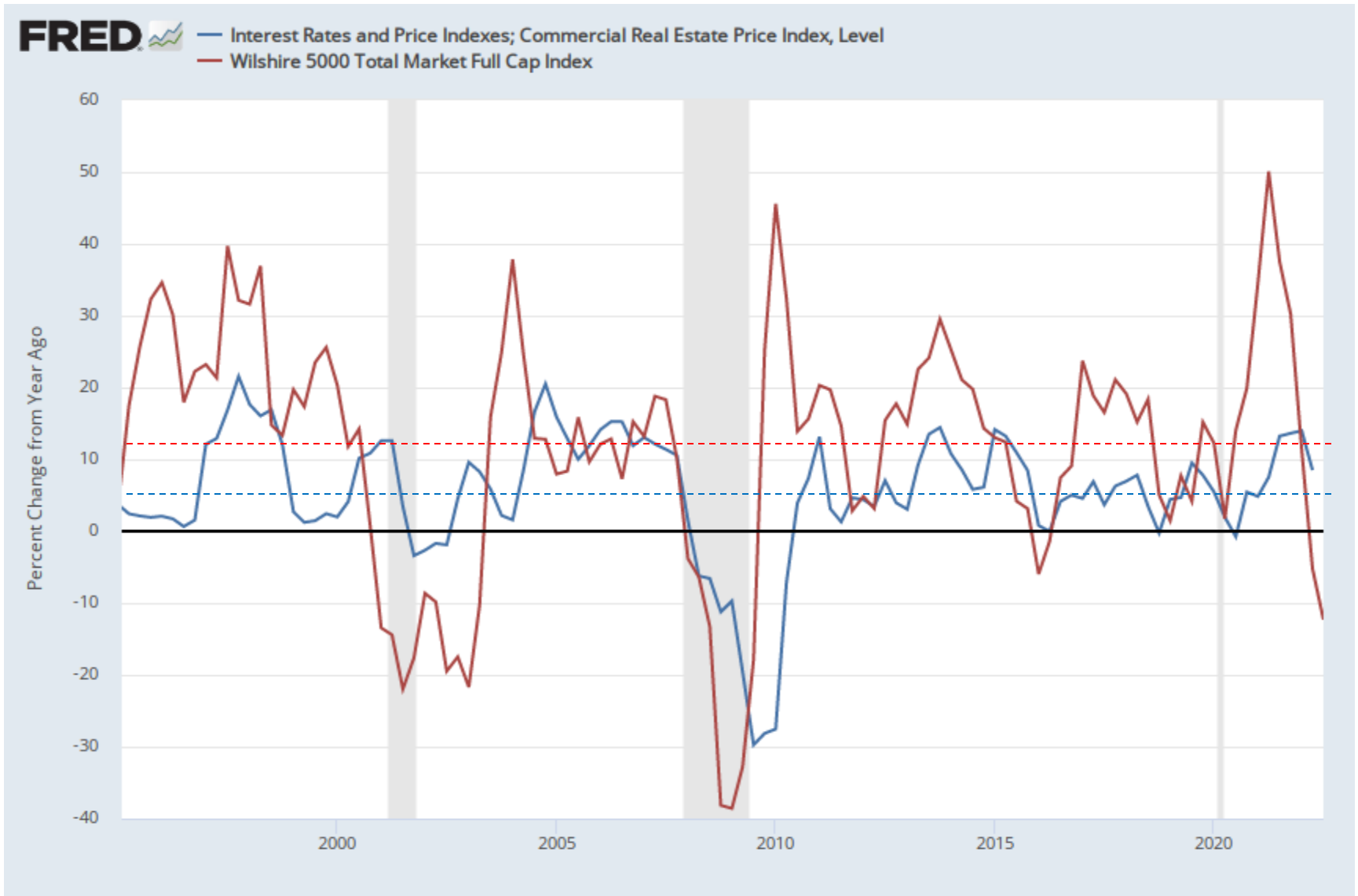
Commercial real estate price index



Year-over-year price changes

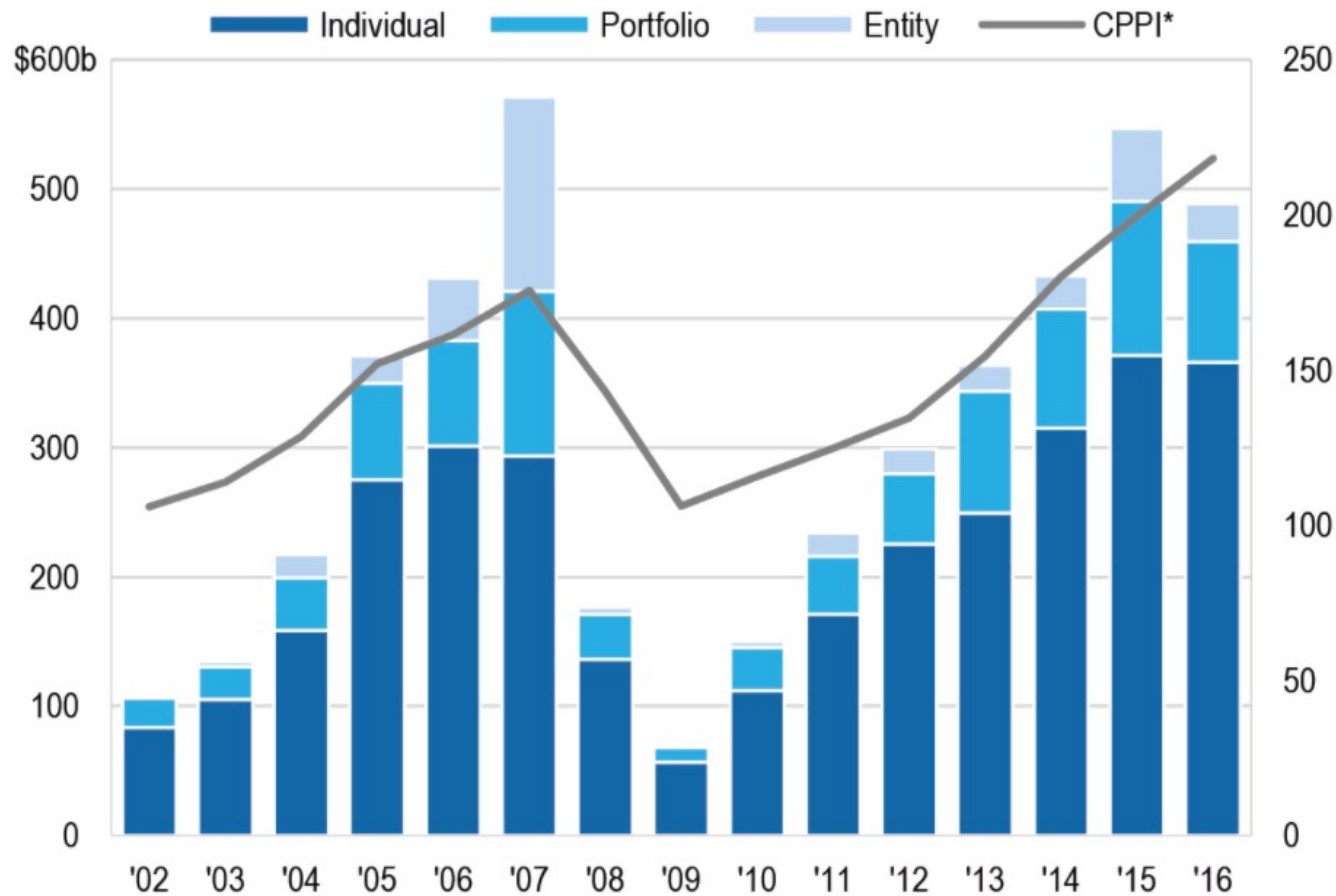


Comparison to stock price changes



CRE trading collapses in downturns

2016 Transaction Volume Dips, Prices Climb

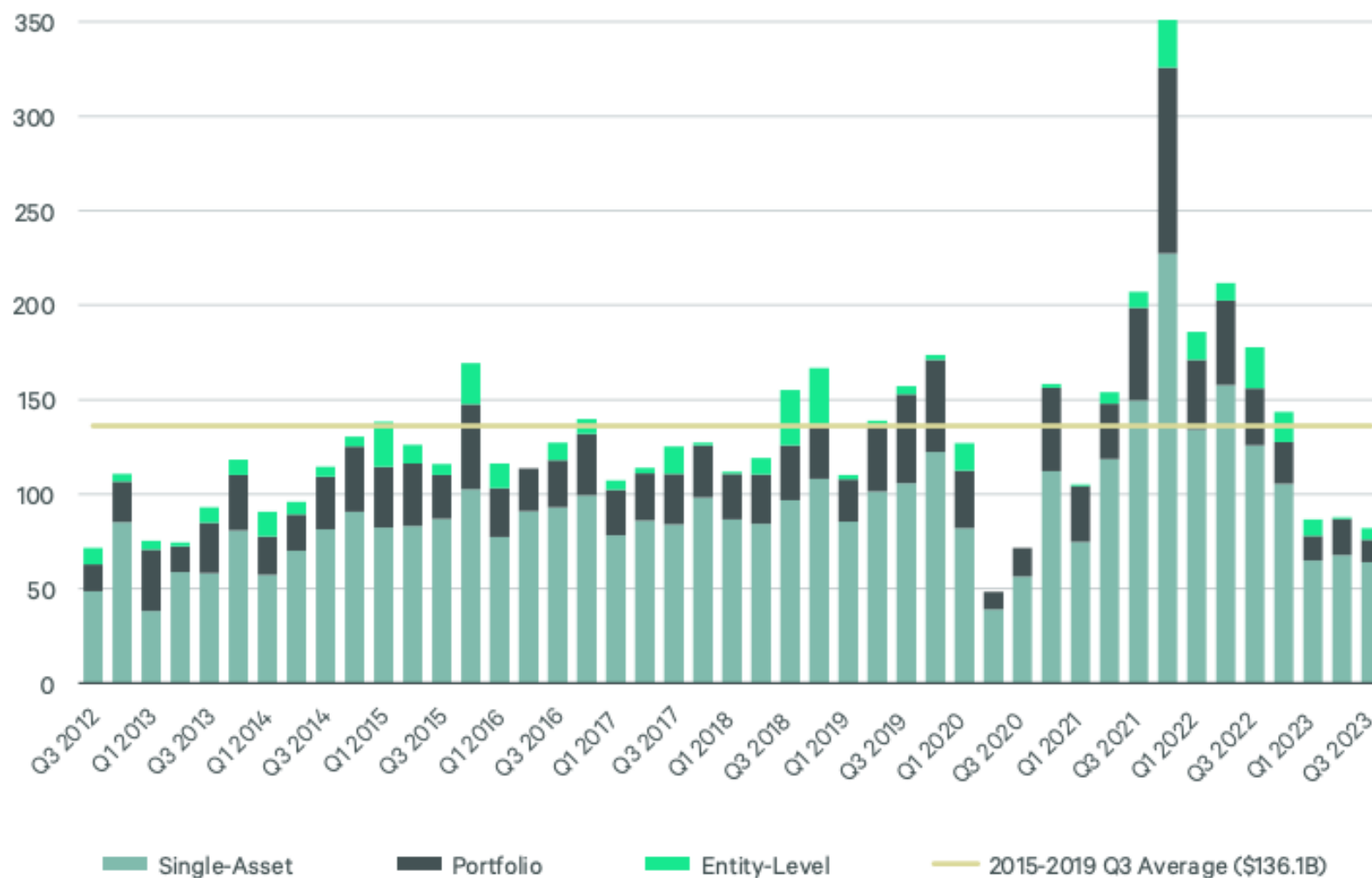


Source: Real Capital Analytics, Moody's

*Moody's/RCA CPPI Q4'16 preliminary

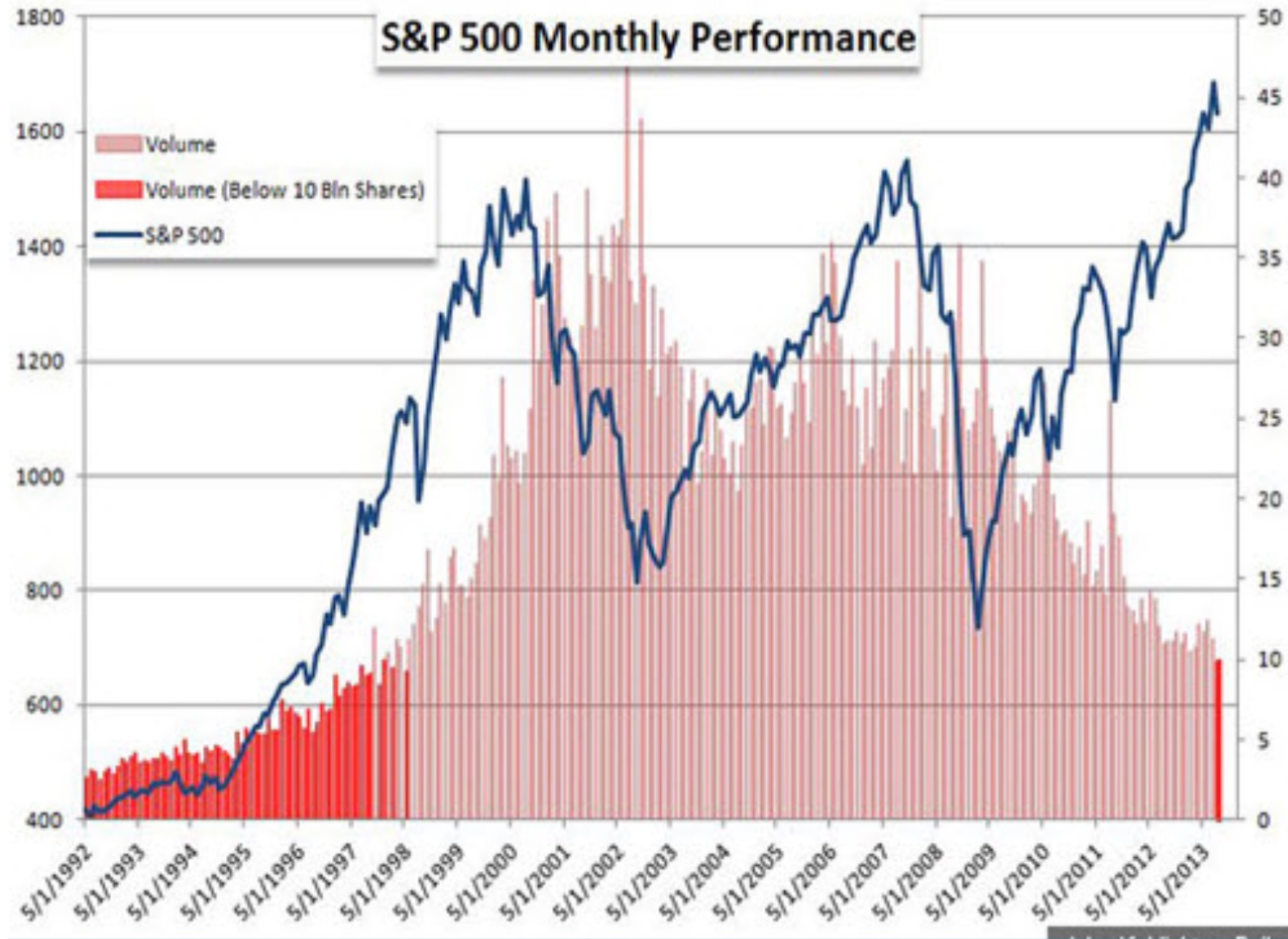
Volume down again this year

Investment Volume (\$ Billions)



Note: Volume excludes development site transactions.
Source: CBRE Research, MSCI Real Assets, Q3 2023.

Stock volume up in downturns



Case study – Hospitality and Housing

Commercial real estate today

- Values down 10+% since March 2022
 - Discount rates up: opportunity cost of investing in real estate has increased, with bond yields rising 3+ p.p.
 - Banks tightening credit supply
- Fundamental distress in office sector
 - Remote work has reduced space demand 20+% and raised vacancy rates 10% pre-COVID to 16% today
- Uncertainty and disagreement
 - Public market REIT valuations down 20+%
 - Property transaction volume down 50%

Summary

- Real estate is economically important
- Applying basic economic analysis of supply and demand can be relevant and enlightening
- Key differences from financial assets include illiquidity, transactions costs and relevance of local, private information